

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Developed and optimized **3D game simulations** using **Panda3D** and **Python** for AI training applications, enhancing rendering fidelity by **15%**.
- Implemented and debugged **Python** and **C++** codebases integrated with **Panda3D**, reducing simulation load times by **20%**.
- Collaborated with multidisciplinary teams to transform AI training requirements into technical solutions, delivering **3 major simulation modules** on schedule.
- Managed code repositories and contributed to version control workflows using **GitHub**, facilitating seamless project collaboration across a remote team of **5 developers**.
- Ensured **cross-platform compatibility** and optimized performance across diverse hardware configurations, achieving consistent frame rates within **10% variance** on target devices.
- Documented architectural decisions, code, and processes, producing comprehensive guides that improved onboarding time for new team members by **30%**.

SKILLS

Game Development – Panda3D, 3D Graphics Pipelines, Simulation Logic, Asset Optimization

Programming Languages – Python, C++

Version Control – GitHub, Code Management, Branching, Pull Requests

Development Methodologies – Agile, Scrum, Code Reviews, Technical Documentation

Platforms & Performance – Cross-Platform Development, Performance Optimization, Hardware Compatibility

AI & Simulation – AI Training Applications, Immersive Simulations

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

AI Training Simulation Engine (HCL Technologies)

- Engineered a bespoke **AI training simulation engine** using **Panda3D** and **Python**, capable of rendering complex 3D environments for machine learning model evaluation.
- Integrated **C++** modules for performance-critical physics calculations, achieving a **40% reduction** in processing time compared to pure Python implementations.
- Implemented advanced **GitHub** workflows including feature branching and pull request reviews, supporting a **distributed team of 8 engineers** across 3 time zones.

- Optimized rendering pipelines and 3D assets, ensuring smooth execution on a range of hardware, from high-end workstations to **legacy systems**, maintaining average frame rates above **60 FPS**.

Financial Transaction Monitoring Dashboard (HCL Technologies)

- Developed and maintained a **payments operations dashboard** supporting a financial-services client, focusing on **transaction monitoring** and operational alerting for payment message failures.
- Implemented data validation and reconciliation workflows using **SQL** (PostgreSQL) and **Python** scripting, reducing manual reconciliation efforts by **25%**.
- Managed production support for the dashboard, responding to P1/P2 incidents within defined SLAs and ensuring **99.9% uptime** for critical monitoring services.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)